

ResearchRadar

Academic Paper Semantic Search & Recommendation System

Discover scientific literature using semantic similarity.

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Source code & notebook: <https://github.com/shahriar-tamjid/research-radar>

5,183

scientific papers in the SciFact corpus

300

test queries scored against official qrels

0.699

NDCG@10 for the hybrid retriever

A compact, end-to-end Information Retrieval and NLP project that compares transparent lexical retrieval (TF-IDF) with dense semantic retrieval (sentence embeddings), blends the two in a simple hybrid retriever, reuses the same embedding space for content-based paper recommendation, and evaluates everything against the **BEIR SciFact** benchmark.

Text → Representation → Similarity → Ranking → Recommendation → Evaluation

At a glance

TF-IDF baseline	Unigrams + bigrams, 50,000 features, sublinear TF
Semantic retrieval	all-MiniLM-L6-v2, 384-dimensional normalised embeddings
Hybrid retrieval	$0.7 \times \text{semantic} + 0.3 \times \text{lexical}$, after per-query min-max normalisation
Recommendation	Content-based, using nearest neighbours in embedding space
Evaluation	Precision@5, Recall@5, MRR@10, NDCG@10 plus qualitative error analysis
Environment	Google Colab, no training, no vector database, no paid API

ResearchRadar is a retrieval and ranking system. It is not a chatbot and not a generative-AI application.

1. Overview

Scientific search is more than matching the exact words in a query. The same idea can be expressed with different terminology, abbreviations or scientific language. ResearchRadar explores this problem by comparing two retrieval approaches: **TF-IDF** for transparent lexical matching and **sentence embeddings** for semantic similarity. A simple **hybrid retriever** then combines the two signals, and the same embedding space is reused to provide **content-based paper recommendations**.

All three retrieval methods are evaluated against the **official SciFact test relevance judgements (qrels)**, so the project delivers both a working demonstration and measurable IR evaluation. The architecture is deliberately small enough to understand, run and explain in a technical discussion.

Key features

• Semantic search over the BEIR SciFact corpus	• TF-IDF lexical retrieval baseline
• Sentence-transformer document and query embeddings	• Cosine-similarity ranking
• Content-based academic paper recommendation	• Simple hybrid lexical + semantic retrieval
• Precision@5, Recall@5, MRR@10 and NDCG@10	• Qualitative error analysis using the qrels
• PCA embedding-space visualisation	• Interactive Plotly embedding explorer
• Interactive Colab search and recommendation demo	• Embedding caching and a fixed random seed

2. System Architecture

The system has two user-facing paths, **semantic search** and **paper recommendation**, which share the same embedding model and similarity machinery. Visual exploration and retrieval evaluation sit on top of both.

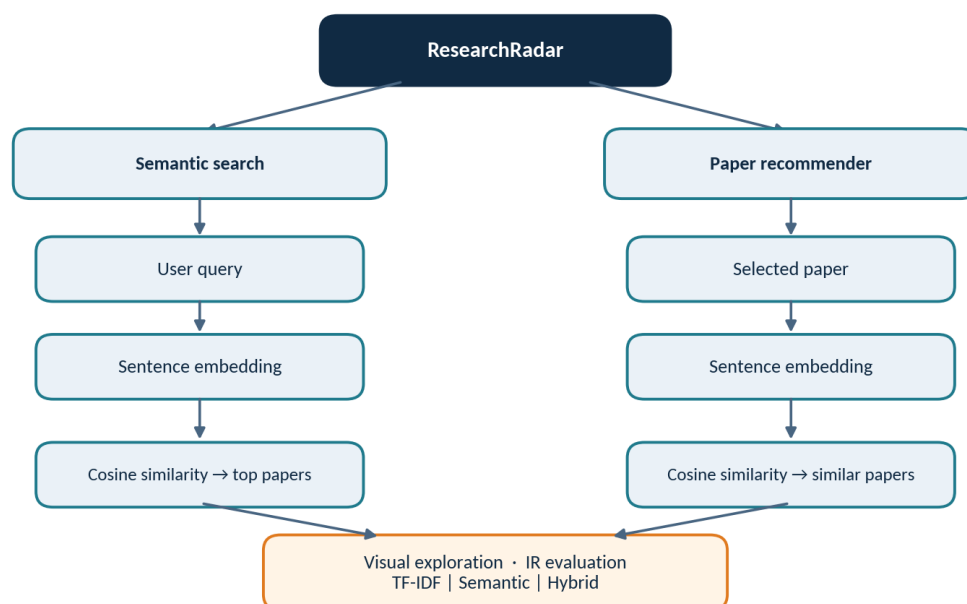


Figure 1. ResearchRadar architecture: two paths over one shared embedding space.

3. NLP and Information Retrieval Concepts

Concept	What it does	Setting used
TF-IDF	Transparent lexical baseline: up-weights terms that distinguish documents, down-weights terms common across the corpus.	English stop words removed; unigrams and bigrams; sublinear TF; 50,000 features
Sentence embeddings	Convert text into dense vectors so that related meaning sits nearby in the embedding space.	sentence-transformers/all-MiniLM-L6-v2; 384 dimensions; normalised; no fine-tuning
Cosine similarity	Scores query/document or document/document pairs by vector alignment.	With normalised embeddings it reduces to a dot product
Content-based recommendation	Ranks the corpus by similarity to a selected paper's embedding.	Not collaborative filtering and not personalised
Hybrid retrieval	Blends semantic and lexical scores after per-query min-max normalisation.	$\alpha = 0.7$ (a demonstration setting, not a tuned optimum)

```
cos(A, B) = (A · B) / (||A|| ||B||)
hybrid_score =  $\alpha$  × semantic_score + (1 -  $\alpha$ ) × lexical_score,  $\alpha = 0.7$ 
```

Interpreting scores. A higher cosine similarity means two vectors are more closely aligned. It is not a probability of relevance, and it does not establish scientific correctness.

4. Dataset and Exploratory Analysis

ResearchRadar uses **SciFact** through the **BEIR** benchmark framework. The official test retrieval split is loaded with BEIR's GenericDataLoader, and title and abstract are joined into a single retrieval text (`combined_text = title + ". " + abstract`).

Structure	Meaning	Recorded run
corpus	Document ID → title and abstract	5,183 documents
queries	Query ID → scientific claim	300 test queries
qrels	Query ID → relevant document ID → grade	300 queries with qrels

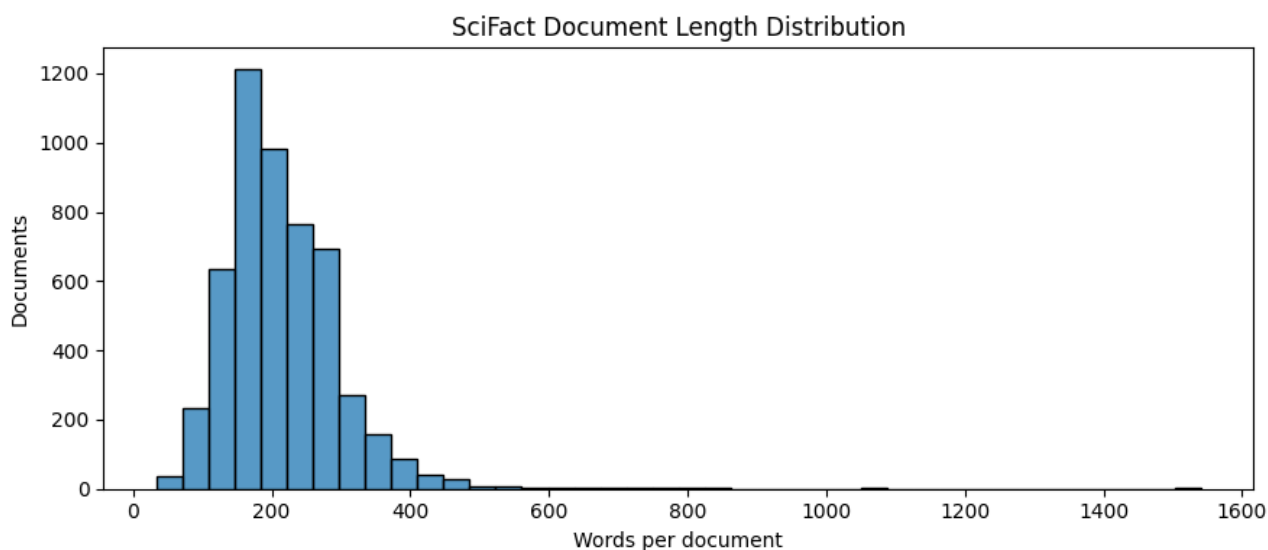


Figure 2. Document length distribution (words per document), generated by the notebook.

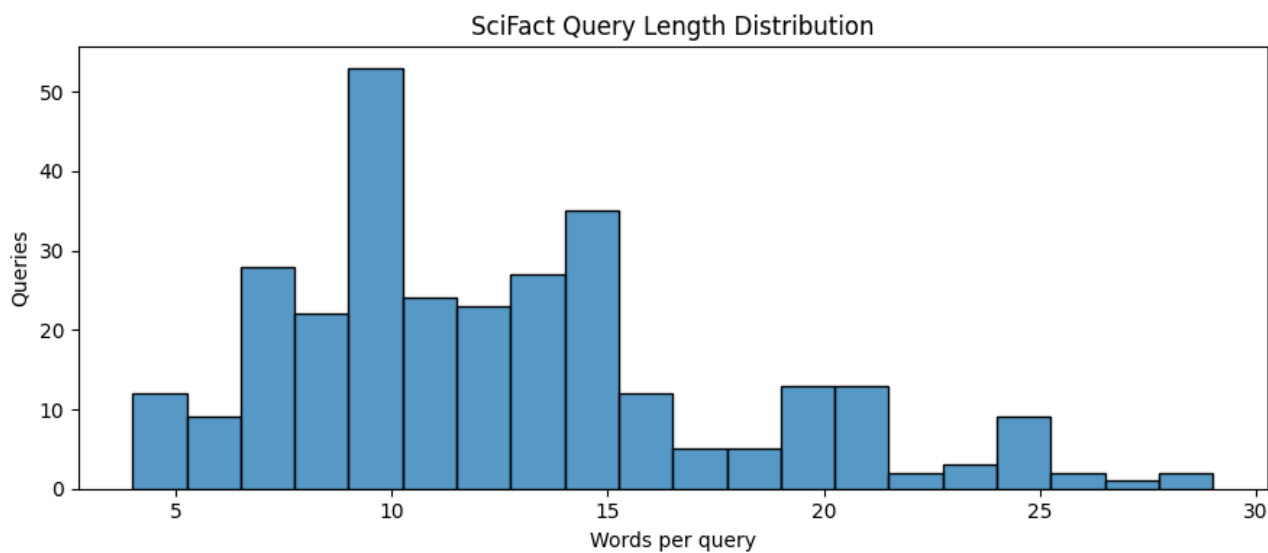


Figure 3. Query length distribution (words per query), generated by the notebook. Queries are far shorter than documents.

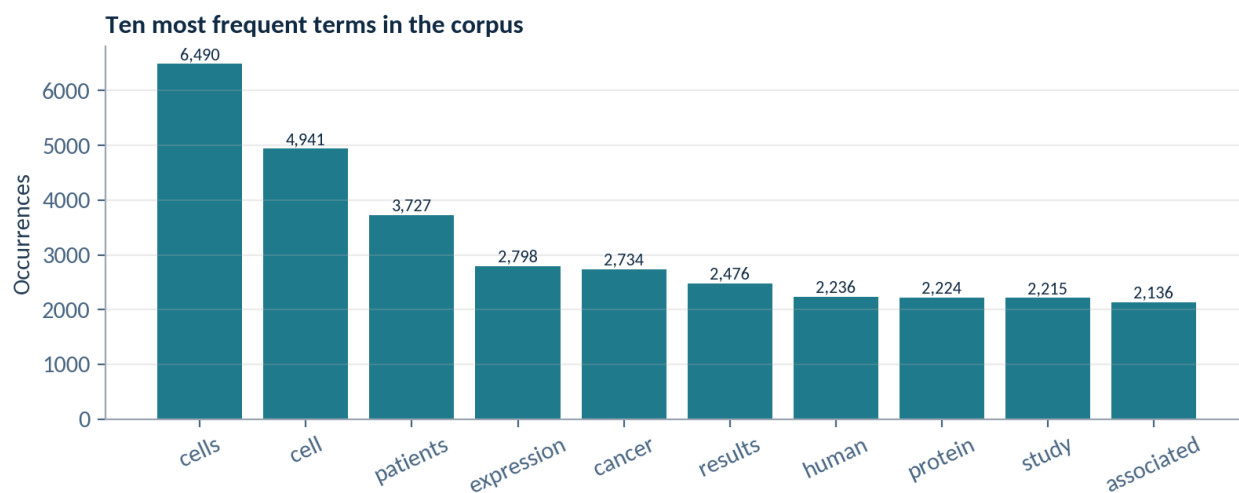


Figure 4. Most frequent corpus terms from the notebook's term-frequency table.

5. Methodology

#	Stage	What happens
1	Dataset loading	SciFact is downloaded into the Colab runtime and the test split is loaded with BEIR.
2	Text preparation	Title and abstract are combined. No aggressive stemming or lemmatisation is applied to the semantic text.
3	Lexical retrieval	A TF-IDF matrix is built for the full corpus ($5,183 \times 50,000$). Queries are projected into the same space and ranked by cosine similarity.
4	Semantic retrieval	Each document is encoded once with all-MiniLM-L6-v2 in batches of 64 (matrix shape $5,183 \times 384$) and cached. Queries are encoded at search time.
5	Document similarity	A selected document's embedding is compared against every other document, with the document itself removed from the candidates.
6	Recommendation	The document-similarity function is reused directly as the recommender, keeping the system transparent.
7	Hybrid retrieval	TF-IDF and semantic scores are min-max normalised per query and combined with $\alpha = 0.7$.
8	Evaluation	All three methods are scored over the full 300-query test set against the official qrels.
9	Error analysis	Queries are grouped by whether the top TF-IDF and top semantic result is qrel-relevant.

6. Retrieval and Recommendation in Action

Same query, three retrievers

For the query “*What is the relationship between inflammation and Alzheimer’s disease?*”, the three methods rank the corpus differently. Titles are shortened as in the notebook output.

Rank	TF-IDF	Semantic	Hybrid
1	Local neuroinflammation and the progression of ...	Local neuroinflammation and the progression of Alzheimer’s disease	Local neuroinflammation and the progression of ...
2	Potential use of γ -secretase modulators in the ...	Chronic inflammation (inflammaging) and its potential contribution to age-associated diseases	Alzheimer’s disease: the two-hit hypothesis.
3	Apolipoprotein E: high-avidity binding to beta ...	Alzheimer’s disease: the two-hit hypothesis.	Genome-wide association study identifies varia...
4	A rare mutation in UNC5C predisposes to late-o...	Immunoproteasome and LMP2 polymorphism in aged and Alzheimer’s disease brains.	Immunoproteasome and LMP2 polymorphism in aged ...
5	Genome-wide association study identifies varia...	Alzheimer’s Disease Risk Gene CD33 Inhibits Microglial Uptake of Amyloid Beta	Apolipoprotein E: high-avidity binding to beta ...

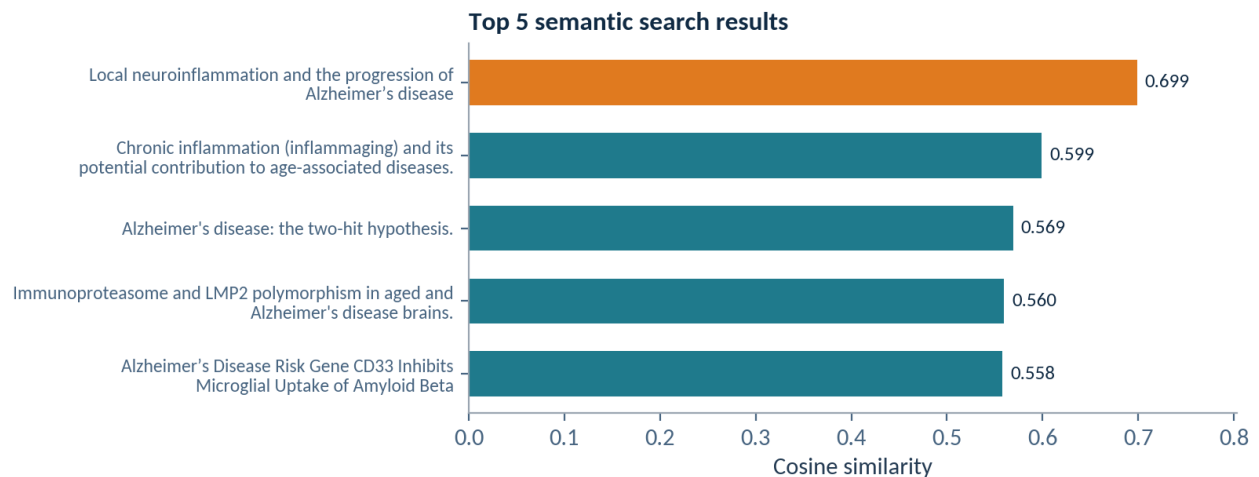


Figure 5. Top five semantic search results with cosine similarity scores.

Content-based recommendation

In the sleep-related demo, the selected paper was “*Cognitive behavioral therapy vs zopiclone for treatment of chronic primary insomnia in older adults: a randomized controlled trial.*” The recommender returned its five nearest neighbours in embedding space.

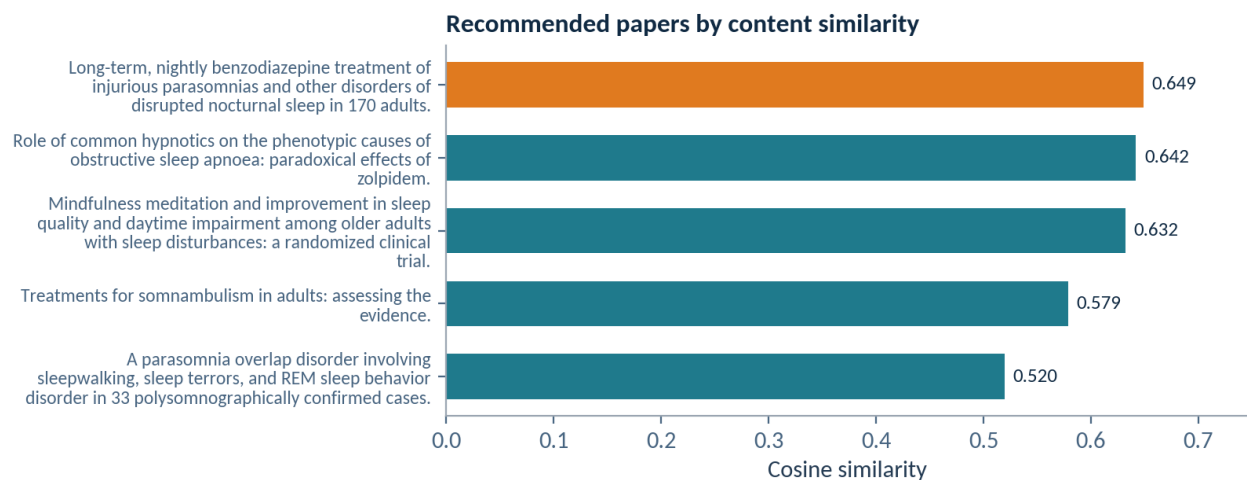


Figure 6. Recommended papers by content similarity for the selected insomnia paper.

These recommendations represent embedding-space similarity, not scientific equivalence or evidence that one paper supports another.

7. Quantitative Evaluation

The metrics: **Precision@5** is the share of the top five that is relevant; **Recall@5** is how much of the known relevant set is retrieved in the top five; **MRR@10** rewards placing the first relevant result high; **NDCG@10** rewards relevant results near the top and accounts for graded relevance.

Method	Precision@5	Recall@5	MRR@10	NDCG@10
TF-IDF	0.1460	0.6758	0.5478	0.5890
Semantic	0.1647	0.7413	0.6068	0.6484
Hybrid	0.1720	0.7766	0.6569	0.6990

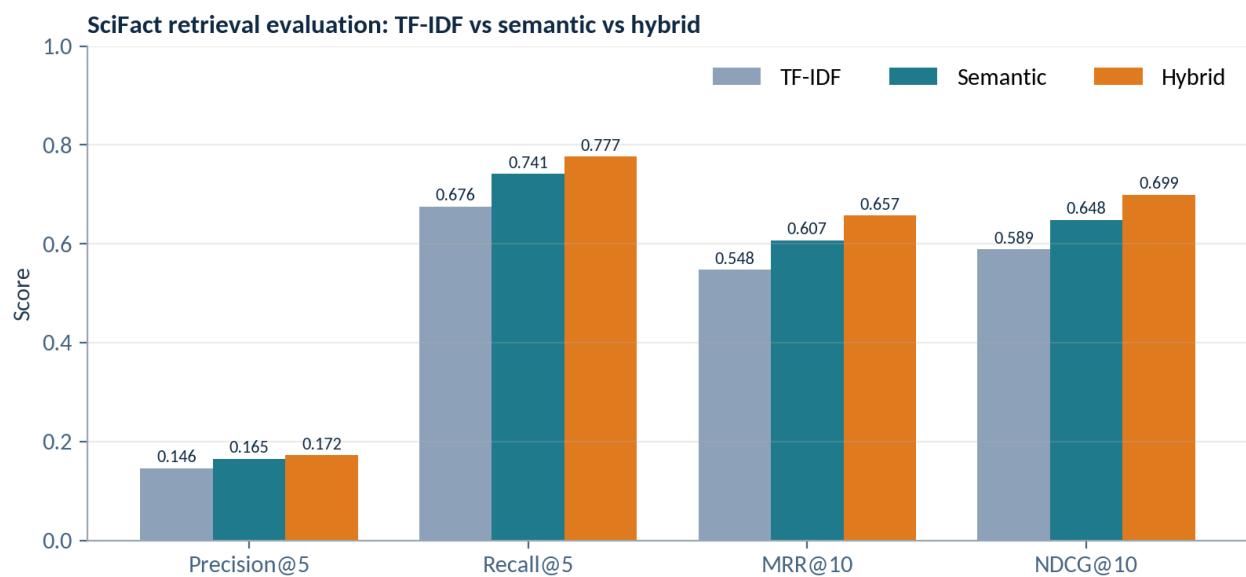


Figure 7. Grouped comparison of the three retrieval methods across all four metrics.

In this recorded run, the hybrid configuration produced the highest value on all four metrics, with semantic retrieval ahead of the TF-IDF baseline throughout. Relative to TF-IDF, hybrid NDCG@10 rises from 0.5890 to 0.6990, and Recall@5 from 0.6758 to 0.7766.

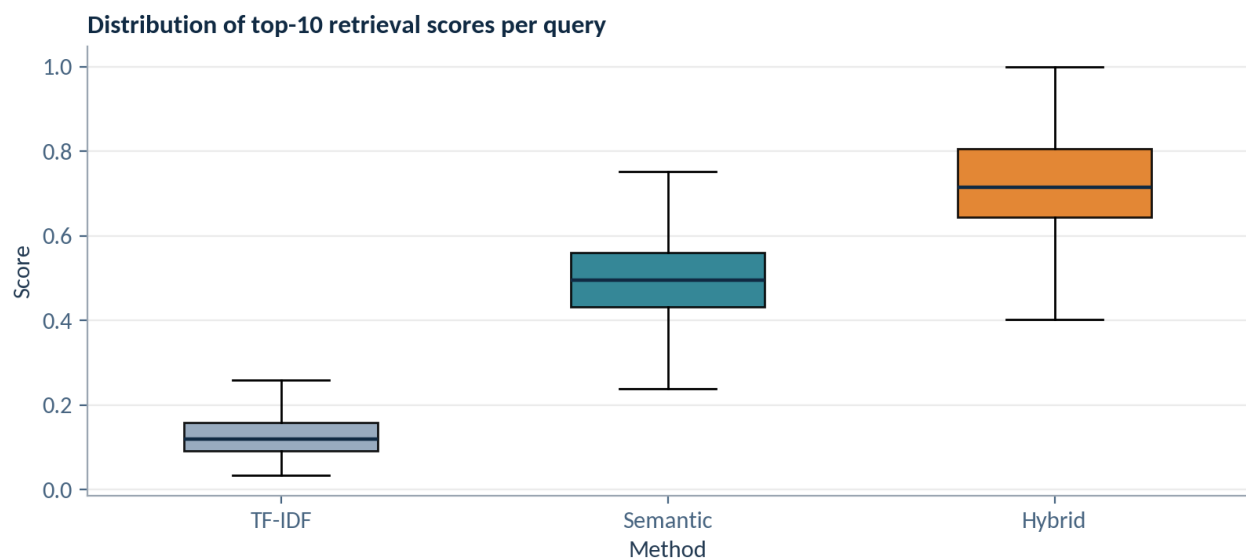


Figure 8. Distribution of top-10 retrieval scores per query (3,000 scores per method).

Read with care. These numbers are results of this particular SciFact experiment, not a universal claim about retrieval methods. The value $\alpha = 0.7$ is a transparent demonstration setting, not a tuned optimum.

8. Error Analysis

Metrics show *how much* performance changes. Error analysis shows *where* the methods behave differently, by checking whether the top TF-IDF and top semantic result is relevant according to the qrels.

Case	Query	Observed behaviour
TF-IDF relevant, semantic not	Activation of PPM1D suppresses p53 function.	TF-IDF ranked a qrel-relevant PPM1D paper first; semantic retrieval chose a different p53-related paper.
Semantic relevant, TF-IDF not	ADAR1 binds to Dicer to cleave pre-miRNA.	Semantic retrieval ranked the relevant ADAR1/Dicer paper first; TF-IDF chose another Dicer-related paper.
Both not relevant	0-dimensional biomaterials show inductive properties.	Neither method returned the known relevant document at rank 1.
Both relevant	AIRE is expressed in some skin tumors.	Both methods returned the same qrel-relevant paper at rank 1.

Category (top-1 result)	Queries
TF-IDF relevant / Semantic not relevant	30
Semantic relevant / TF-IDF not relevant	48
Both not relevant	119
Both relevant	103

Lexical overlap and semantic similarity can fail or succeed for different reasons. The notebook treats these as observations from the benchmark rather than causal explanations.

9. Embedding-Space Exploration

Up to 750 document embeddings are projected from 384 dimensions to two principal components. The notebook presents this as an interactive Plotly scatter plot with the document ID and title on hover; the static view is shown here. PC1 explains 8.04% and PC2 3.97% of the variance; this is a 2D projection for exploration and should not be read as preserving the full structure of the original 384-dimensional space.

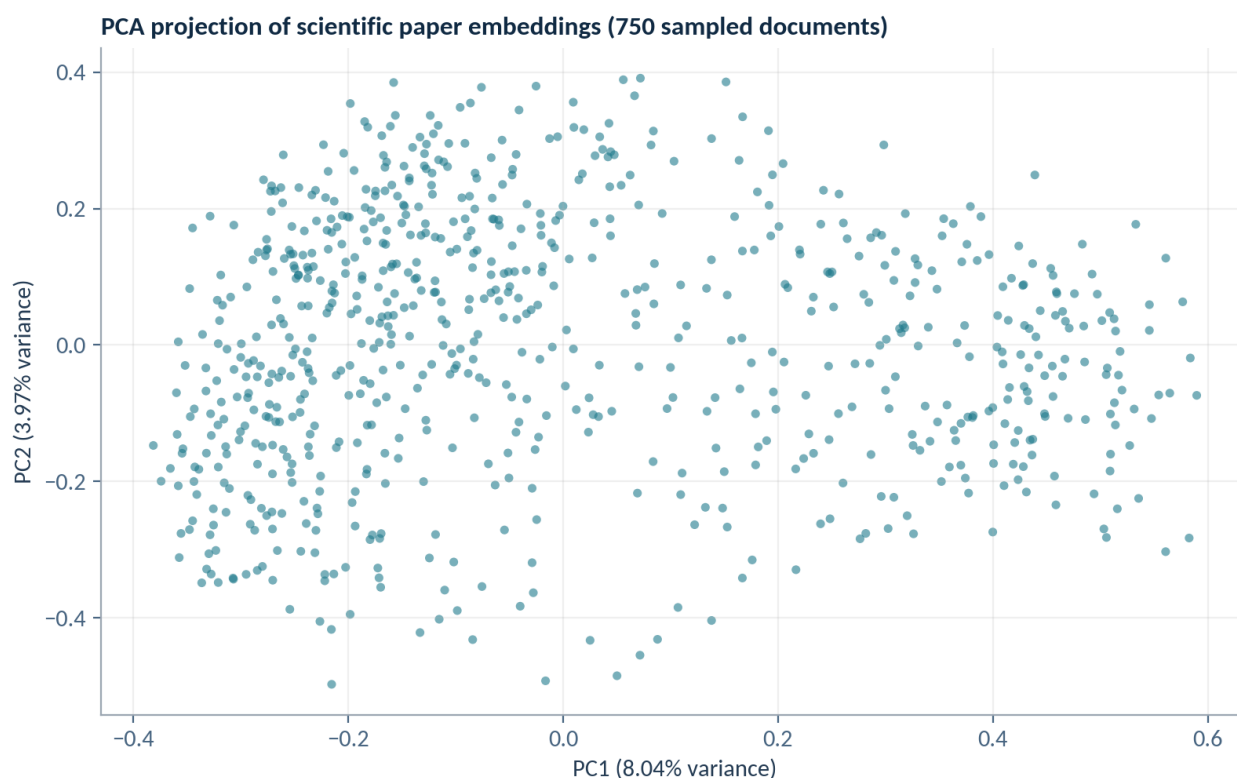


Figure 9. PCA projection of 750 sampled paper embeddings (PC1 + PC2 explain 12.02% of variance).

10. Interactive Demo and Example Queries

The notebook includes an ipywidgets interface for Google Colab: enter a research question, view the top five semantic results, select a paper, and request similar papers. It calls the same retrieval and recommendation functions used throughout the notebook rather than a separate hidden system. Eight demonstration queries run against the real SciFact corpus; results are not hard-coded.

- 1 What is the relationship between sleep deprivation and cognitive performance?
- 2 Does physical activity reduce the risk of cardiovascular disease?
- 3 How does inflammation affect Alzheimer's disease?
- 4 What factors influence cancer progression?
- 5 Can machine learning improve medical diagnosis?
- 6 How are biomarkers used in disease detection?
- 7 What is the role of oxidative stress in human disease?
- 8 Does diet influence the development of cardiovascular disease?

11. Reproducibility, Limitations and Next Steps

Reproducibility and performance

- Random seed fixed at 42 for Python, NumPy and PyTorch where available.
- Batched embedding generation (batch size 64) with normalised embeddings and vectorised similarity.

- Local .npy embedding cache, validated against model name, corpus size, embedding dimension and document IDs.
- No model training, transformer fine-tuning, external vector database or paid API required.

Limitations

- SciFact is a relatively small benchmark compared with modern scholarly search indexes.
- The embedding model is a general-purpose encoder, not fine-tuned on SciFact.
- Semantic similarity does not guarantee scientific correctness; a retrieval ranking is not an evidence verdict.
- Recommendations are content-based and not personalised: there is no user history or collaborative filtering.
- The corpus lacks citations, authors, years and live indexing, and the metrics are benchmark-specific.

Future improvements

BM25 as an additional lexical baseline · scientific-domain embedding models · cross-encoder reranking · citation-graph-based recommendations · metadata filtering by year, topic, author or venue · personalised recommendation · larger scientific retrieval datasets · multimodal paper retrieval · explainable retrieval · systematic hyperparameter tuning for the hybrid retriever.

12. Conclusion

ResearchRadar shows, clearly and concretely, how to move from **raw scientific text to representations, similarity scores, ranked retrieval, recommendations, visualisation and empirical evaluation** in a single reproducible Colab project. On the recorded SciFact run, combining lexical and semantic signals gave the strongest results of the three methods tested.

Core	Python, Google Colab, NumPy, Pandas, scikit-learn, SciPy
NLP / IR	BEIR, SciFact, Sentence Transformers, PyTorch, TF-IDF, cosine similarity
Visualisation	Matplotlib, Seaborn, Plotly, ipywidgets
Utility	tqdm, JSON/path-based local caching

Explore the project. Notebook, README and requirements are available on GitHub:

<https://github.com/shahriar-tamjid/research-radar>. Open ResearchRadar . ipynb in Google Colab and run it top to bottom; SciFact downloads automatically.

References

Thakur, N. et al. *BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models*. NeurIPS Datasets and Benchmarks Track. github.com/beir-cellar/beir

Sentence Transformers ([sbnet.net](https://www.sbert.net)) and the pretrained model [sentence-transformers/all-MiniLM-L6-v2](https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2) (huggingface.co/sentence-transformers/all-MiniLM-L6-v2). SciFact is used as a benchmark through BEIR; the project does not claim ownership of the dataset.